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B.Tech. Degree IV Semester Examination April 2019

ME 404 APPLIED THERMODYNAMICS (2006 Scheme)

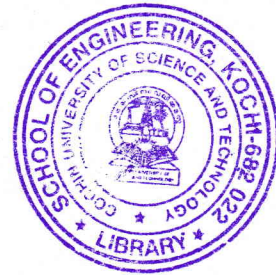
Time: 3 Hours

Maximum Marks: 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) State and explain Carnot theorem.
- (b) Explain the Clausius inequality.
- (c) What do you understand by PVT surface?
- (d) Steam at 10 bar pressure and 0.9 dryness fraction is cooled at constant volume at 160°C. What will be its final condition?
- (e) Explain Dalton's law of partial pressure.
- (f) List the types of air compressors.
- (g) Explain Bomb calorimeter with a neat sketch.
- (h) Explain enthalpy of combustion.



PART B

(4 × 15 = 60)

- II. (a) Define and explain availability as applied to thermodynamics. (5)
- (b) A reversible heat engine is supplied 900 kJ of heat from a heat source at 500 K. The engine develops 300 kJ of net work and rejects heat to two heat sinks at 400 K and 300 K. Determine the engine thermal efficiency and the magnitude of heat interaction with each of the sink. (10)

OR

- III. A closed system contains 2 kg of air and during an adiabatic expansion process, there occurs a change in its pressure from 500 kPa to 100 kPa and its temperature from 350 K to 320 K. If the volume doubles during the process, calculate: (15)
 - (i) the maximum work
 - (ii) the change in availability
 - (iii) irreversibility

- IV. A steam power plant operates on a theoretical reheat cycle. Steam at 25 bar pressure and 400°C is supplied to the high pressure turbine. After its expansion to dry state, the steam is reheated at constant pressure to its original temperature. Subsequent expansion occurs in the low pressure turbine to a condenser pressure of 0.04 bar. Considering feed pump work, determine: (15)
 - (i) quality of steam at entry to condenser
 - (ii) thermal efficiency
 - (iii) Specific steam consumption

OR

(P.T.O.)

- V. (a) Explain any one high pressure boiler with a neat sketch. (5)
 (b) A boiler generates steam at 10 bar and 200°C from feed water at 37°C. (10)
 The coal used in the furnace has a calorific value of 33965 kJ/kg and 10 kg of steam is generated per kg of coal burnt. Calculate:
 (i) equivalent evaporation from and at 100°C
 (ii) boiler power

- VI. A mixture of 1 kg of oxygen and 2 kg of nitrogen occupies 1.2 m³ volume (15)
 at temperature 300 K. Assuming perfect gas behaviour, determine:
 (i) specific volume
 (ii) pressure
 (iii) gas constant
 (iv) molecular mass

OR

- VII. Write short notes on: (15)
 (i) Amagat's law of partial volume
 (ii) Advanced problem on adiabatic mixing
 (iii) Rotary compressors

- VIII. A hydrocarbon fuel of unknown composition is burned with air and an (15)
 Orsat analysis of the composition of products gives

$$\text{CO}_2 = 7.26\% \quad \text{CO} = 2.42\% \quad \text{Oxygen} = 7.5\%$$

Determine:

- (i) Empirical formula of the fuel
 (ii) Composition of the fuel on mass basis
 (iii) Percent theoretical air used for combustion

OR

- IX. (a) Differentiate between gravimetric and volumetric analysis. (10)
 (b) Explain the adiabatic flame temperature. (5)
